SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**BALA.S**
Register Number	**192524427**

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I BALA S(192524427) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: bala
Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1.A media company wants to migrate its video streaming platform to the cloud.The system must handle millions of concurrent users with minimal buffering.Design a cloud architecture using scalability and caching techniques.Explain how load balancing and auto-scaling can ensure smooth streaming.Discuss cost optimization strategies for such high-demand workloads

Scalability and Caching Techniques

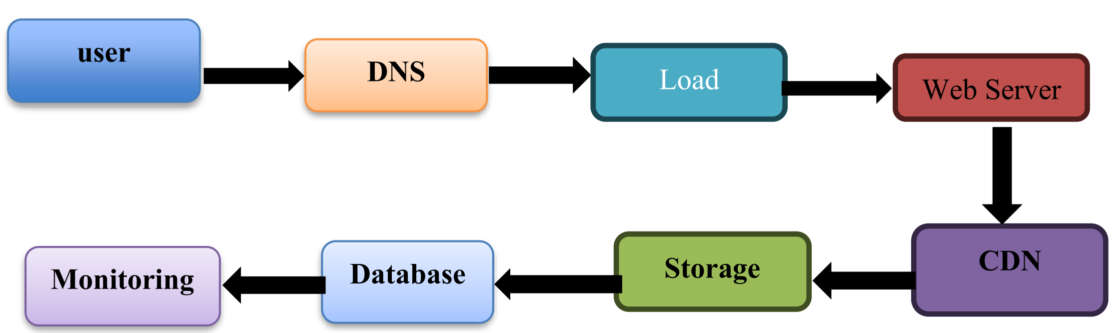
- Content Delivery Network (CDN): Stores video content at edge locations close to users
- Caching: Frequently watched videos and metadata are cached using services
- Object Storage: Store videos in scalable cloud storage (e.g., Amazon S3, Azure Blob Storage).
- Microservices: Separate services for authentication

Load Balancing

- Distributes incoming user requests across multiple streaming servers.
- Prevents any single server from becoming overloaded.
- Performs health checks and redirects traffic away from failed servers.
- Improves availability and response time.

Auto-Scaling

- Automatically increases the number of servers during peak traffic.
- Reduces servers when demand decreases.
- Ensures consistent streaming performance with minimal buffering.
- Optimizes resource usage without manual intervention.



2.A fintech startup is deploying applications using containers across multiple cloud providers.They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.

Proposed Solution

- Deploy applications using Docker containers and manage them with Kubernetes across multiple cloud providers (such as AWS, Azure, and Google Cloud).
- Use a centralized CI/CD pipeline to automate deployment and updates.

Container Orchestration and Multi-Cloud Strategy

- Use Kubernetes to deploy, manage, and monitor containers.
- Package applications in Docker containers for consistency across clouds.
- Use Helm Charts or Terraform for consistent infrastructure deployment.
- Implement centralized monitoring and logging across all cloud providers.

Service Discovery

- Kubernetes automatically assigns DNS names to services.
- Pods communicate using service names instead of IP addresses.
- Failed containers are automatically replaced without affecting users.

Scaling

- Horizontal Pod Autoscaler (HPA) automatically increases or decreases pods based on CPU or memory usage.
- Cluster Autoscaler adds or removes worker nodes according to workload.
- Load balancers distribute traffic evenly among healthy containers.

Benefits of Multi-Cloud Deployment

- High Availability: Applications remain available even if one cloud provider fails.
- Disaster Recovery: Easy backup and failover across clouds.
- Avoid Vendor Lock-in: Freedom to switch or combine cloud providers.

Risks of Multi-Cloud Deployment

- Increased management complexity.
- Higher networking and data transfer costs.
- Security and compliance become more challenging.
- Different cloud services may have compatibility issues.
- Monitoring and troubleshooting are more difficult across multiple platforms

3.An e-learning platform experiences data loss due to accidental deletion in the cloud.The organization needs a robust backup and disaster recovery strategy.Design a solution using cloud-native backup, replication, and versioning.Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.Provide steps to ensure business continuity during failures.

Cloud-Native Backup, Replication, and Versioning:

- Automated Backups: Schedule daily full backups and frequent incremental backups of databases.

- Cross-Region Replication: Replicate data to another cloud region for disaster recovery.

- Object Versioning: Keep multiple versions of files so deleted or modified files can be restored easily.

- Snapshots: Create regular storage snapshots for quick recovery

Recovery Time Objective (RTO):

- Use standby servers and automated failover.

- Restore applications from snapshots and backups quickly.

- Load balancers redirect users to the disaster recovery site.

Recovery Point Objective (RPO):

- Perform frequent backups and continuous data replication.
- Versioning minimizes data loss by allowing restoration of recent file versions.
- Continuous synchronization ensures only a few minutes of data may be lost.

Steps to Ensure Business Continuity:

- Enable automated backups for databases and storage.
- Turn on object versioning to recover deleted files.
- Configure cross-region replication for critical data.
- Maintain a disaster recovery site in another region.
- Use load balancers and auto-scaling to maintain service availability.
- Regularly test backup restoration and disaster recovery procedures.
- Continuously monitor systems and configure alerts for failures.

Benefits:

- Protects against accidental deletion.
- Ensures high availability and fast recovery.
- Reduces downtime and data loss.
- Supports uninterrupted learning services during failures

4.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Hybrid Cloud Design:

- Store sensitive data and critical applications in the on-premises data center
- Connect both environments using a secure private connection

Secure Communication:

- Use VPN or dedicated private connections for secure communication.
- Protect data in transit using TLS/SSL encryption.
- Configure firewalls and network security groups to restrict unauthorized access.
- Monitor network traffic using security monitoring tools.

Identity Management:

- Implement Identity and Access Management (IAM) with Single Sign-On (SSO).
- Enable Multi-Factor Authentication (MFA) for all users.
- Synchronize on-premises directory services with cloud IAM.

Encryption

- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.
- Rotate encryption keys regularly.

Challenges in Hybrid Cloud Management

- Increased complexity in managing both on-premises and cloud environments.
- Maintaining consistent security and compliance policies.
- Network latency between environments.

5.A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Data Replication

- Use synchronous replication for critical data (low latency regions).
- Use asynchronous replication for distant regions to reduce delays.
- Replicate databases, storage, and backups across regions.

Failover Mechanism

- Configure health checks on application servers.
- If Region A fails, the global load balancer automatically redirect traffic to Region B.
- Enable automatic database failover to the replica.
- Regularly test disaster recovery and failover procedures.

Monitoring and Alerting

- Continuously monitor CPU, memory, storage, network, and application performance.
- Monitor service health, database replication status, and response times.

Benefits

- High availability with minimal downtime.
- Automatic recovery during regional failures.
- Improved performance by serving users from the nearest region.
- Better disaster recovery and business continuity.
- Increased reliability and customer satisfaction.